

Predicting Wildfire Risk via MLP, CNN, and Bayesian Modeling

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Project



Background & Purpose

- Climate change is significantly amplifying wildfire risk, with longer and more active fire seasons⁷
- Previous literature has explored these models with some gaps:
 - Lack of MLP inclusion in model benchmarking^{3,4}
 - Accuracy of current CNNs can be improved^{5,6}
 - Environmental predictors/socioeconomic factors drawn from satellite data as input to Bayes LR → images not used directly^{1,2}

Purpose: Accurately predict areas at risk of wildfire from satellite images

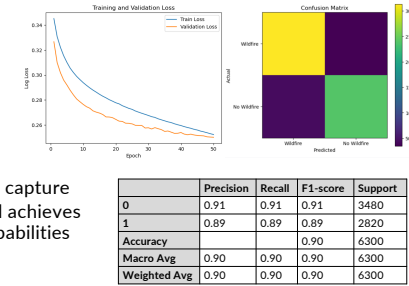
Goal: Directly compare three machine learning models

- Multilayer perceptron (MLP)
- Convolutional neural network (CNN)
- Bayesian logistic regression (Bayes LR)

Results

Multilayer Perceptron

- Test accuracy 90% with balanced performance across classes
- Model is slightly better at predicting wildfire-risk regions
- Though MLP cannot capture spatial structure, still achieves strong predictive capabilities



Conclusion

	Precision	Recall	F1-score	Accuracy
MLP	0.90	0.90	0.90	0.90
CNN	0.94	0.94	0.94	0.94
Bayesian LR	0.84	0.83	0.84	0.82

- CNN outperformed MLP (accuracy of ~94%, compared to 90%), demonstrating the value of kernels in extracting relevant features
- As expected, MLP and CNN far outperformed the Bayes LR for binary image classification
- Bayes LR was most lacking, though the image data inherently doesn't lend itself to prior/posterior distributions
- Both MLP and Bayesian LR use each pixel as independent features → loss of spatial context → weaker compared to CNN

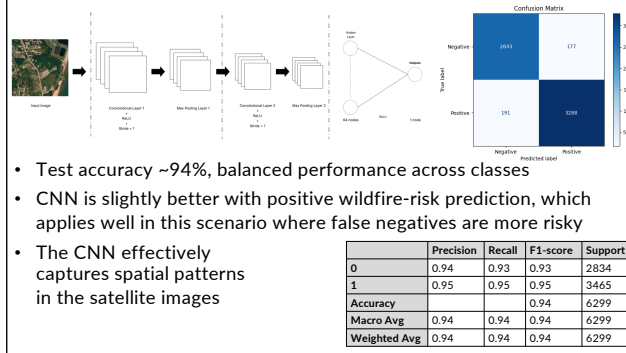
Methodology

Dataset & Pre-processing

- 350x350px labeled Canadian satellite images⁸, pulled from locations that previously experienced wildfires (> 0.01 acres burned)
- Data labeled as at risk for wildfire (1) or not at risk (0) (55%, 45% per class, resp.)
- 70/15/15 used as train/valid/test split
- Pre-process with image resizing and pixel normalization between [0, 1]



Convolution Neural Network



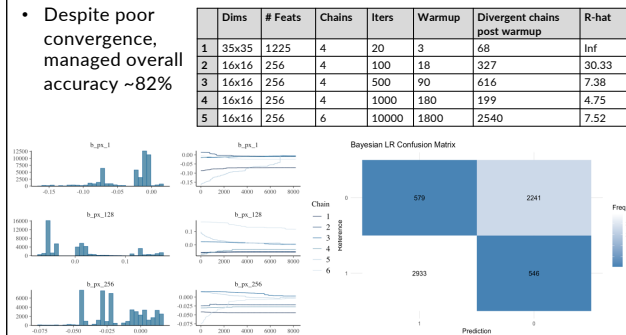
- Test accuracy ~94%, balanced performance across classes
- CNN is slightly better with positive wildfire-risk prediction, which applies well in this scenario where false negatives are more risky
- The CNN effectively captures spatial patterns in the satellite images

Model Setup

MLP (Python, manual)	2-layer MLP (1 hidden layer with ReLU activation) - Optimize with binary cross-entropy loss by gradient descent - Sigmoid for output probability
CNN (Python, Keras, TensorFlow)	2 convolutional layers that apply a ReLU activation, (3,3) kernel size, stride of 1 16 units, 32 units respectively - Max pooling applied after each convolutional layer with a (2,2) kernel size - Pooled outputs are flattened for hidden layer - Hidden layer applies ReLU, outputting 64 units - Output layer applies Sigmoid, generate single unit
Bayesian LR (R, brms)	- RGB of each pixel combined into scalar, flattened to vector for input → each pixel represents a feature - Bernoulli prior and posterior 4 chains, 1000 iterations, 18% burn-in

Bayesian Logistic Regression

- After resizing in preprocessing, several hyperparameters modified
- Decreased dimensions and increased iterations seems to improve chain mixing (R-hat ~ 1.00 = good mixing)
- Many features and memory/time constraints prevents convergence



Future Research

- Use CNN kernels to reduce the dimensions/extract features from images → input to Bayesian model to improve performance
- Integrating temporal data to the data → models might be able to detect wildfire risk earlier on, capture patterns over time
- Broadening the dataset to include different climates → a more generalized model with broader applications
- Account for temperature levels and vegetation index → improve Bayes performance
- Integrate a Satellite API to visualize model performance more clearly
- Work with organizations like the Environmental Protection Agency to update the model with new data in real-time

References

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